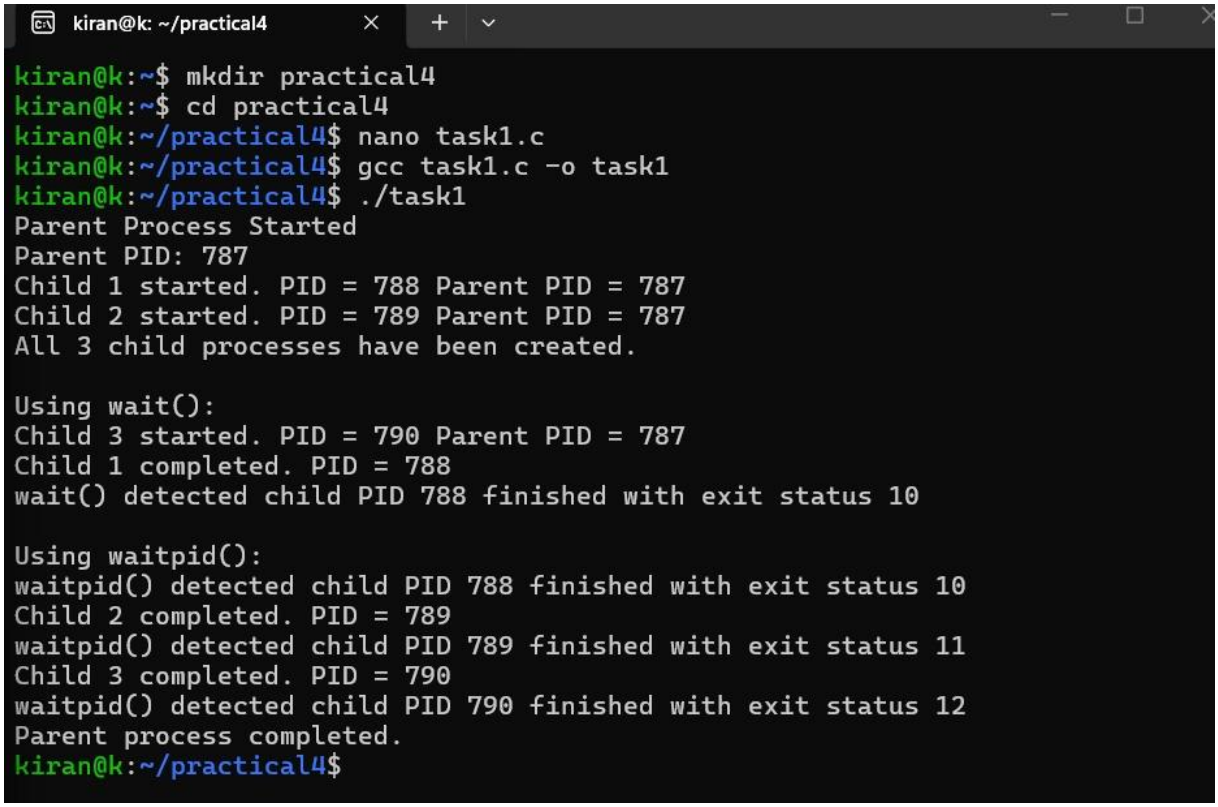
Practical Week-4

Task 1: Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions.



```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid[3];
    int status;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    for (int i = 0; i < 3; i++)
    {
        pid[i] = fork();

        if (pid[i] == 0)
        {
            printf("Child %d started. PID = %d Parent PID = %d\n",
                   i + 1, getpid(), getppid());

            sleep(i + 1);

            printf("Child %d completed. PID = %d\n",
                   i + 1, getpid());

            exit(10 + i);
        }
    }

    printf("All 3 child processes have been created.\n");

    printf("\nUsing wait():\n");

    pid_t finished = wait(&status);

    printf("wait() detected child PID %d finished with exit status %d\n",
           finished, WEXITSTATUS(status));

    printf("\nUsing waitpid():\n");
```

```
    for (int i = 0; i < 3; i++)
    {
        waitpid(pid[i], &status, 0);

        printf("waitpid() detected child PID %d finished with exit status %d\n",
               pid[i], WEXITSTATUS(status));
    }

    printf("Parent process completed.\n");

    return 0;
}
```

Task 2: Create a scenario where a child process becomes a zombie process. Investigate the process table and then modify the program to eliminate zombie processes using proper synchronization techniques.

```
kiran@k:~/practical4$ nano task2.c
kiran@k:~/practical4$ gcc task2.c -o task2
kiran@k:~/practical4$ ./task2
Parent process: PID = 837
Child PID = 838
Parent sleeping...
Child process: PID = 838
Child process terminating...
Parent terminating...
kiran@k:~/practical4$ ./task2 &
[1] 852
Parent process: PID = 852
Child PID = 854
Parent sleeping...
Child process: PID = 854
Child process terminating...
kiran@k:~/practical4$ ps -el
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S     0       1       0  0  80   0 -  6045 -      ?        00:00:01 system
5 S     0       2       1  0  80   0 -   795 -      hvc0     00:00:00 init-s
0 S     0       6       2  0  80   0 -   795 -      hvc0     00:00:00 init
4 S     0      49       1  0  79  -1 - 12606 -      ?        00:00:00 system
4 S   989      84       1  0  80   0 -  5635 -      ?        00:00:00 system
4 S     0      90       1  0  80   0 -  8779 -      ?        00:00:01 system
4 S     0     147       1  0  80   0 -   722 -      ?        00:00:00 chrony
4 S     0     151       1  0  80   0 -  1118 -      ?        00:00:00 cron
4 S   995     155       1  0  80   0 -  2204 -      ?        00:00:00 dbus-d
4 S     0     166       1  0  80   0 - 11023 -      ?        00:00:00 networ
4 S     0     173       1  0  80   0 -  4610 -      ?        00:00:00 system
4 S   100     179       1  0  80   0 - 55137 -      ?        00:00:00 rsyslo
4 S     0     237       1  0  80   0 -  1373 -      tty1     00:00:00 agetty
4 S   988     239     147  0  80   0 -  5105 -      ?        00:00:00 chrony
4 S     0     252       1  0  80   0 - 30864 -      ?        00:00:00 unatte
1 S   988     260     239  0  80   0 -  3023 -      ?        00:00:00 chrony
4 S     0     344       2  0  80   0 -  2162 -      ?        00:00:00 login
4 S  1000     397       1  0  80   0 -  5582 -      ?        00:00:00 system
5 S  1000     399     397  0  80   0 -  5727 -      ?        00:00:00 (sd-pa
4 S  1000     424     344  0  80   0 -  1534 poll_s pts/1    00:00:00 bash
5 S     0     696       2  0  80   0 -   796 -      ?        00:00:00 Sessio
5 S     0     697     696  0  80   0 -   800 -      ?        00:00:00 Relay(
4 S  1000     704     697  0  80   0 -  1567 do_wai pts/0    00:00:00 bash
0 S  1000     852     704  0  80   0 -   692 hrtime pts/0    00:00:00 task2
1 Z  1000     854     852  0  80   0 -     0 -      pts/0    00:00:00 task2
0 R  1000     863     704  0  80   0 -  1798 -      pts/0    00:00:00 ps
kiran@k:~/practical4$ Parent terminating...
```

```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child process: PID = %d\n", getpid());

        printf("Child process terminating...\n");

        exit(0);
    }
    else
    {
        printf("Parent process: PID = %d\n", getpid());

        printf("Child PID = %d\n", pid);

        printf("Parent sleeping...\n");

        sleep(30);

        printf("Parent terminating...\n");
    }

    return 0;
}
```

```
kiran@k:~/practical4$ nano task2.c
kiran@k:~/practical4$ gcc task2.c -o task2
kiran@k:~/practical4$ ./task2
Parent process: PID = 837
Child PID = 838
Parent sleeping...
Child process: PID = 838
Child process terminating...
Parent terminating...
kiran@k:~/practical4$ ./task2 &
[1] 852
Parent process: PID = 852
Child PID = 854
Parent sleeping...
Child process: PID = 854
Child process terminating...
kiran@k:~/practical4$ ps -el
F S   UID     PID    PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S     0       1       0  0  80   0 -  6045 -      ?        00:00:01 system
5 S     0       2       1  0  80   0 -   795 -      hvc0     00:00:00 init-s
0 S     0       6       2  0  80   0 -   795 -      hvc0     00:00:00 init
4 S     0      49       1  0  79  -1 - 12606 -      ?        00:00:00 system
4 S   989      84       1  0  80   0 -  5635 -      ?        00:00:00 system
4 S     0      90       1  0  80   0 -  8779 -      ?        00:00:01 system
4 S     0     147       1  0  80   0 -   722 -      ?        00:00:00 chrony
4 S     0     151       1  0  80   0 -  1118 -      ?        00:00:00 cron
4 S   995     155       1  0  80   0 -  2204 -      ?        00:00:00 dbus-d
4 S     0     166       1  0  80   0 - 11023 -      ?        00:00:00 networ
4 S     0     173       1  0  80   0 -  4610 -      ?        00:00:00 system
4 S   100     179       1  0  80   0 - 55137 -      ?        00:00:00 rsyslo
4 S     0     237       1  0  80   0 -  1373 -      tty1     00:00:00 agetty
4 S   988     239     147  0  80   0 -  5105 -      ?        00:00:00 chrony
4 S     0     252       1  0  80   0 - 30864 -      ?        00:00:00 unatte
1 S   988     260     239  0  80   0 -  3023 -      ?        00:00:00 chrony
4 S     0     344       2  0  80   0 -  2162 -      ?        00:00:00 login
4 S  1000     397       1  0  80   0 -  5582 -      ?        00:00:00 system
5 S  1000     399     397  0  80   0 -  5727 -      ?        00:00:00 (sd-pa
4 S  1000     424     344  0  80   0 -  1534 poll_s pts/1    00:00:00 bash
5 S     0     696       2  0  80   0 -   796 -      ?        00:00:00 Sessio
5 S     0     697     696  0  80   0 -   800 -      ?        00:00:00 Relay(
4 S  1000     704     697  0  80   0 -  1567 do_wai pts/0    00:00:00 bash
0 S  1000     852     704  0  80   0 -   692 hrtime pts/0    00:00:00 task2
1 Z  1000     854     852  0  80   0 -     0 -      pts/0    00:00:00 task2
0 R  1000     863     704  0  80   0 -  1798 -      pts/0    00:00:00 ps
kiran@k:~/practical4$ Parent terminating...
kiran@k:~/practical4$ nano task2.c
[1]+  Done                    ./task2
kiran@k:~/practical4$ gcc task2.c -o task2
kiran@k:~/practical4$ ./task2
Parent process: PID = 905
Child PID = 906
Child process: PID = 906
Child process terminating...
Child process reaped. No zombie.
kiran@k:~/practical4$ ps -el | grep task2
kiran@k:~/practical4$
```

```
  GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid == 0)
    {
        printf("Child process terminating...\n");

        exit(0);
    }
    else
    {
        wait(NULL);

        printf("Zombie process removed.\n");
    }

    return 0;
}
```